Melbourne Airbnb Market Screening

CMCE30005 Final Project Report | Group 2

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Project repository: https://github.com/maksym-xu/CMCE30005-TheNextChapter/tree/main

1. Introduction

Our client plans to lease homes in Greater Melbourne for Airbnb short stays. Rent and setup costs come before guest income, so a poor lease ties up scarce startup funds.

We recommend City of Melbourne two- and three-bedroom apartments first, then Yarra Ranges three-bedroom houses and townhouses. These groups had the highest shares of busy homes (30+ reviews a year, about 180+ booked nights) and typical estimated revenue of AUD 30,600–48,600 a year before costs, against AUD 24,700 overall.

Within these groups, half of the homes that accepted one-night stays and were priced at or below similar homes were busy (50.4%), against a third of all their homes (33.4%). Section 6 turns this into a search order, a checklist and a cost test.

2. Problem Definition and Objectives

The business question is: **where should a rent-to-rent operator with limited funds search, and which homes should it inspect first?** We set three objectives:

1. Rank property groups by their share of homes with strong guest activity, and express this in nights and revenue.
2. Identify which home settings go with strong activity, and whether they help pick homes within the leading groups.
3. Turn the results into a search order and a four-week inspection plan.

Objective 1 needs only a simple summary by group (Section 5.2). Objective 2 uses a model to separate settings from location, then tests a simpler two-setting rule.

A **group** combines council area, dwelling type and bedroom count. A home is **busy** (reaches our target) when it receives at least 30 guest reviews in the year before data collection, a level reached by a quarter of set-aside reference homes (Section 3). Inside Airbnb (n.d.) assumes half of stays are reviewed and each lasts three nights (or the minimum stay if longer). Estimated nights are capped at 255 a year; 30 reviews means about 180 booked nights.

What review activity measures. Guests review only after a stay, so reviews track completed bookings, a guide to demand and occupancy; this matters because empty nights are the operator's main risk. Review counts show neither profit (rents and costs are unknown) nor satisfaction (reviews can be negative), so we convert them into nights and revenue and check sentiment through star ratings (Section 5.1). We do not rank on revenue: Inside Airbnb estimates it from reviews and advertised prices, and without rents a higher price is not a better lease.

Since the interim report, we replaced its random forest with an explainable logistic regression.

3. Data Description and Preparation

We use one Inside Airbnb snapshot of Melbourne, supplied through the subject LMS and collected 17 June–1 July 2026 (Inside Airbnb, 2026). It holds 25,728 listings and 1,026,690 dated reviews. We did not use the calendar, which cannot separate bookings from blocked nights.

Cleaning converted prices to numbers, standardized missing entries, counted amenities and rebuilt annual review counts from review dates. Price is missing for 6,553 listings (25%) and bedrooms for 4,679 (18%); 4,477 listings (17%) have never been reviewed.

We keep homes priced AUD 30–1,500 that are entire rental units or condos (*apartments*) or entire homes or townhouses (*houses/townhouses*), with one to three bedrooms and a first review at least a year before collection, so each has a full year of reviews.

We set aside about one in five hosts (1,342 homes) to set an independent target: of their 906 homes in the final groups, a quarter received 30 or more reviews (Table 1).

Table 1. From supplied listings to analysis sample

Step	Homes
All listings	25,728
Entire homes with one to three bedrooms	16,576
Apartments, condos, houses and townhouses	14,895
Priced AUD 30–1,500	11,099
First review at least a year before collection	6,891
After setting aside 1,342 reference-host homes	5,549
Analysis sample: 14 groups with at least 50 homes	**3,873**

The 3,873 analysis homes belong to 1,726 hosts in six council areas, the only areas with groups of at least 50 homes. The price rule drops 3,796 residential listings: 77.8% had no review in the past year, against 17.0% of those kept.

4. Methodology

Group comparison. We compare the share of busy homes in the 14 groups; redrawing hosts at random 1,000 times shows how much each share could move by chance.

Two-setting rule. To test whether a simple summary does as well as the model, we split priority-group homes by one-night stays and by price at or below the group median.

Model. Logistic regression turns six inputs into a chance of being busy (R Core Team, 2026); its weights show how each input moves that chance with the others held fixed. Table 2 lists the outcome and inputs.

Table 2. Outcome and inputs of the model

Variable	Definition
Outcome	1 if the home had at least 30 reviews in the year, otherwise 0
Council area	One of six
Dwelling type	Apartment, or house/townhouse
Bedrooms	One, two or three
Price compared with similar homes	Price relative to the group median, per 10%
Accepts one-night stays	Minimum stay of one night (yes or no)
Amenity count	Amenities listed, per ten

We compare this **full model** with two simpler options. The **group rate** (a group's share of busy homes) scores all its homes equally, which within a group is **random picking**. The **reduced model** drops price and minimum stay to show what they add.

Assumptions. Homes of the same host can perform alike, so they stay together in every test. Each 10% price step, or ten more amenities, has the same effect wherever it occurs; Appendix A finds little overlap between inputs and only weak signs of curvature.

Testing. Each of 50 tests builds the model on 70% of hosts and scores the other 30%. In each priority group, each method picks a quarter of these homes, and we compare the share that were busy.

5. Analysis and Results

5.1 Data exploration: most homes are quiet; busy homes accept one-night stays and are well rated

Guest activity is very uneven (Figure 1). The median home received 15 reviews, a quarter (981 homes, 25.3%) received 30 or more, and 334 homes (8.6%) received none.

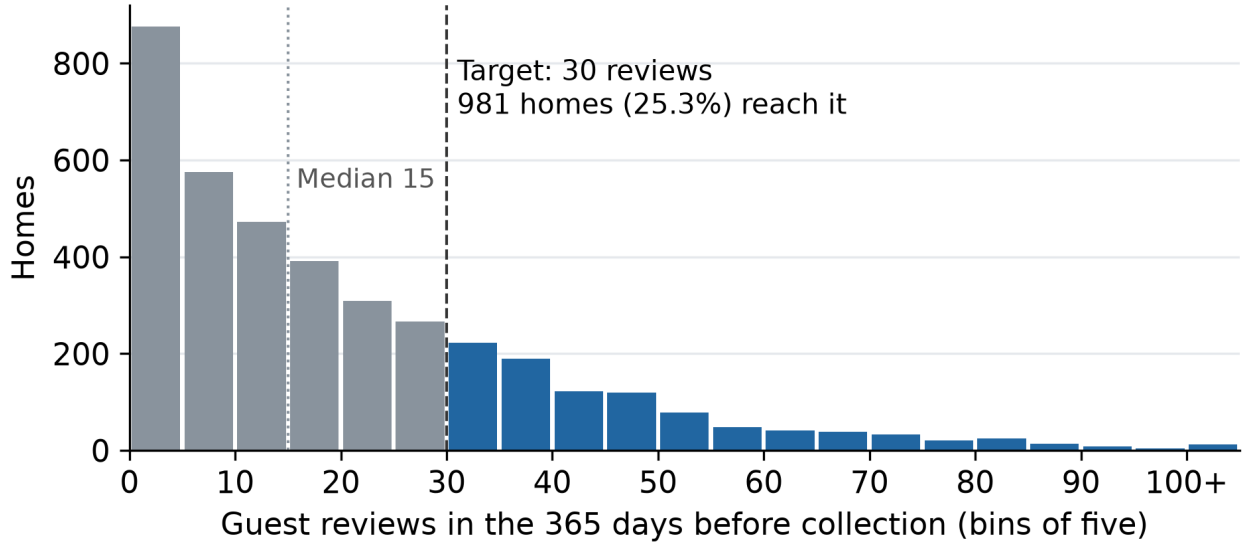


Figure 1. Annual reviews per home, 3,873 analysis homes. Source of all figures and tables: Inside Airbnb (2026); Group 2 analysis.

Table 3. Busy homes compared with the rest

Characteristic	Busy (30+ reviews)	Below 30	All homes
Homes	981	2,892	3,873
Median nightly price (AUD)	254	266	262
Accept one-night stays	67.7%	37.5%	45.1%
In the City of Melbourne	76.9%	59.6%	64.0%
Median estimated nights (cap 255)	255	60	96
Median estimated revenue (AUD a year)	58,400	16,600	24,700
Mean guest rating (out of 5 stars)	4.84	4.69	4.73
Rated below 4.5 stars	1.1%	15.2%	11.6%

Busy homes were slightly cheaper but far more often accepted one-night stays and sat in the City of Melbourne (Table 3); their higher revenue estimate comes from their review counts, not higher prices. They were also better rated: only 1.1% averaged below 4.5 stars, against 15.2% of the rest, though averages over few reviews vary more (Huang & Xu, 2026). So high review counts do not appear to reflect unhappy guests.

5.2 Descriptive comparison: a simple summary puts City apartments first

Group rates range from 8.2% to 36.6% against 25.3% overall, and the three priority groups lead (Figure 2, Table 4).

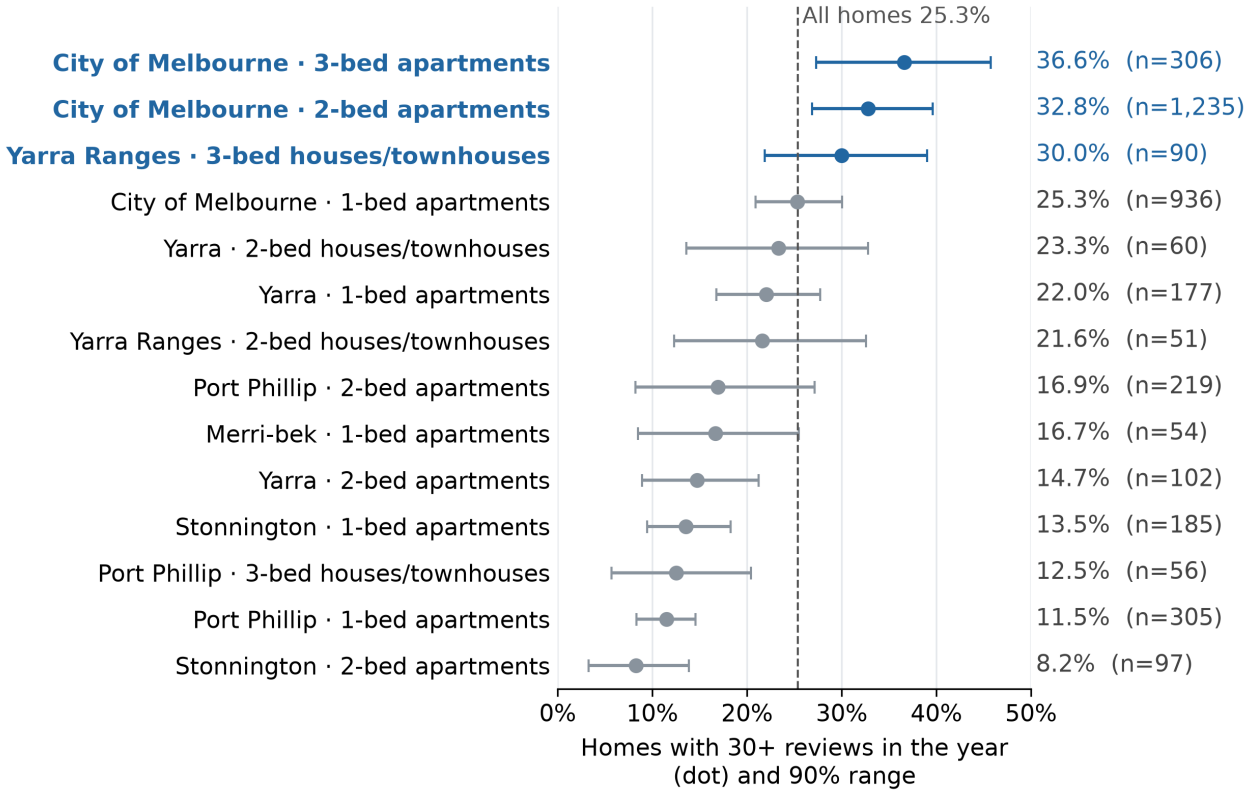


Figure 2. Share of busy homes in all 14 groups, with 90% ranges from redrawing hosts; priority groups in blue.

Table 4. Priority groups: median estimated nights and annual revenue (AUD)

Group	Homes	Share busy	Nights, all homes	Revenue, all homes	Revenue rank	Revenue, below 30
City 3-bed apartments	306	36.6%	126	48,600	1	32,100
City 2-bed apartments	1,235	32.8%	108	30,600	5	16,600
Yarra Ranges 3-bed houses	90	30.0%	120	46,100	2	31,100
All 14 groups	3,873	25.3%	96	24,700	–	16,600

The same three groups lead without the price limits or one-year review rule (Huang & Xu, 2026). In 1,000 host redraws, the City groups stayed in the top three in 94% and 90% of them, and Yarra Ranges houses in only 64%.

When the rankings differ. On median revenue (Table 4), Yarra Ranges houses rank second and City two-bedroom apartments fifth, mainly because the houses charge more a night (AUD 426 against 285). With rents unknown, activity guides the search; revenue tests each lease.

5.3 Predictive analysis: one-night stays and price go with busy homes

Holding other inputs fixed, three settings an operator controls go with busy homes (Figure 3; Table A1). A typical City two-bedroom apartment has a **42%** chance of being busy with one-night stays but **18%** with a two-night minimum or longer. Pricing 20% above similar homes lowers it to **36%**; ten more amenities raise it to **48%**. These are associations: changing a setting alone does not guarantee more bookings.

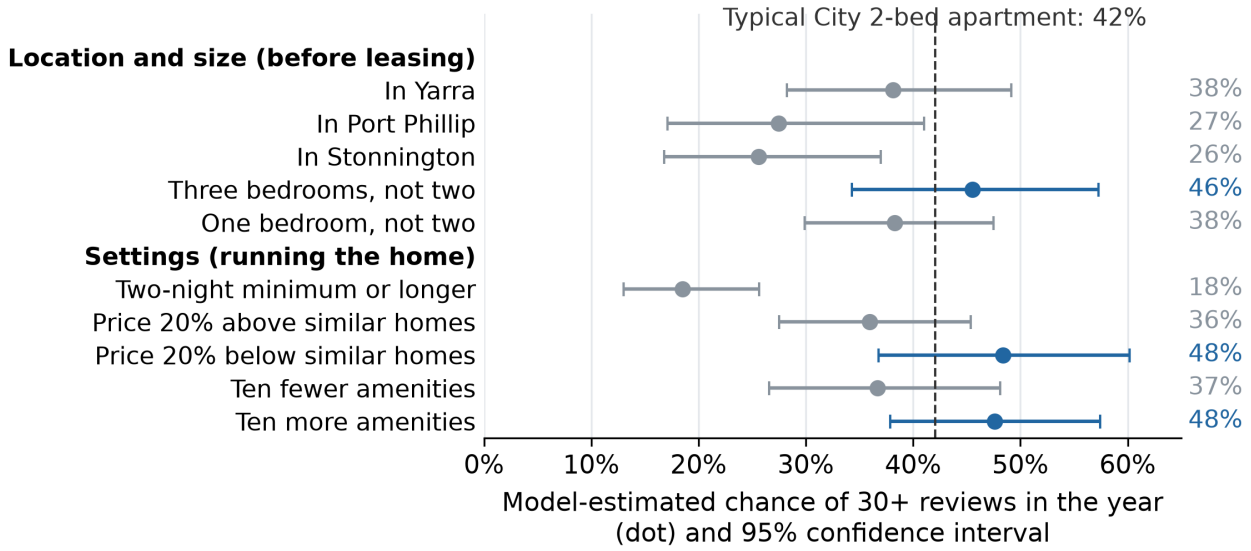


Figure 3. Model-estimated chance that a typical City two-bedroom apartment is busy when one input changes, with 95% confidence intervals.

A simple two-setting rule captures most of this (Table 5). In the priority groups, **50.4%** of homes that accepted one-night stays and were priced at or below their group median were busy, against **33.4%** of all their homes and **9.3%** of homes with neither. Only 20 of 90 Yarra Ranges houses accept one-night stays, so the rule rests mainly on City apartments.

Table 5. Share of busy homes in the three priority groups, by two settings

Minimum stay	Price at or below group median	Price above group median
One night accepted	50.4% (462 homes)	36.1% (477 homes)
Two nights or more	30.2% (358 homes)	9.3% (334 homes)

Tested on unseen hosts, **49.6%** of the model's top quarter in each priority group were busy, against **38.6%** for the reduced model and **32.4%** for random picking (Figure 4); the model beat random picking in 46 of 50 tests. The two-setting rule does about as well (it picks 28% of homes), though it was checked on the homes that suggested it. The model also separates location from settings: with the same settings, the typical apartment's 42% chance would be 26–27% in Port Phillip or Stonnington (Figure 3). Table A2 gives its hits and misses across all 14 groups.

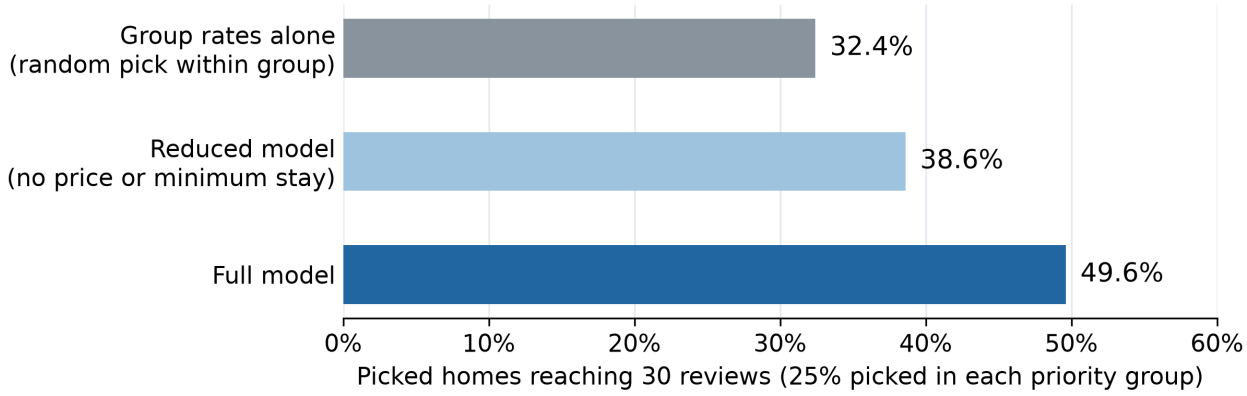


Figure 4. Busy share of picked homes (a quarter of each priority group), averaged over 50 tests.

5.4 Over time: the leading groups stayed ahead

We ranked ten groups (2,657 homes with two years' history) on the earlier year. The priority groups came top: 33.6% of their homes reached 34 reviews (the earlier year's reference level), against 19.9% elsewhere (Figure 5). A year later they still led (**27.1% against 14.9%**) and remained the top three, although Yarra Ranges houses fell to 21.4%.

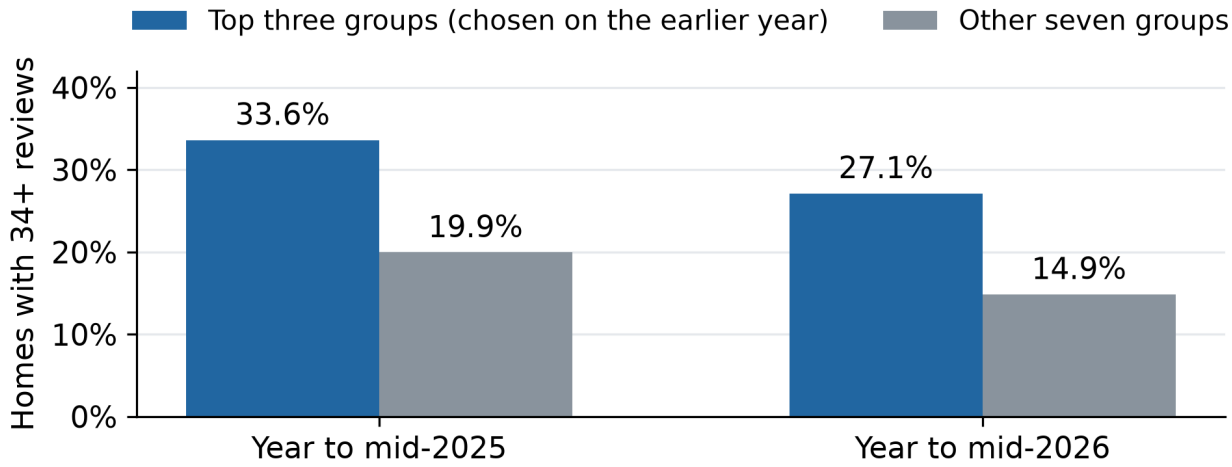


Figure 5. The same 2,657 homes in two consecutive years; target 34 reviews.

6. Key Findings and Recommendations

Key findings

1. **Where:** City of Melbourne two- and three-bedroom apartments led on busy homes, then Yarra Ranges three-bedroom houses, less certainly; all three stayed ahead a year later and had above-median revenue (Figures 2 and 5, Table 4).
2. **What goes with high activity:** One-night stays, a price at or below similar homes and, with other inputs fixed, more amenities (Figure 3). In the priority groups, 50.4% of homes with the first two were busy, against 9.3% with neither (Table 5), though one-night stays partly inflate review counts. Busy homes were also better rated (Table 3).
3. **Which home first:** These two settings found 50.4% busy, near the model's 49.6%, though only the model was tested on unseen hosts; random picking found 32.4% (Table 5, Figure 4).

Recommendations

1. **Search City of Melbourne two- and three-bedroom apartments first, then Yarra Ranges three-bedroom houses** (Figure 2, Table 4). Between leases that pass, choose the larger cash-flow margin; if margins are similar, choose the group with the higher share of busy homes.
2. **Prefer homes that can take one-night stays at a price at or below similar listings** (Table 5, Figure 3). At inspection, check that building rules allow one-night stays and that this price covers costs (Recommendation 3); then trial these settings and track results (Table 6).
3. **Compare weekly rent and running costs with group revenue** (Figure 6). Typical weekly revenue is **AUD 590** for City two-bedroom apartments, **930** for City three-bedroom apartments and **890** for Yarra Ranges houses, but **320, 620 and 600** for homes below target (under 30 reviews). Because these are medians, only half of a group's homes covered a cost equal to its typical figure. Drop candidates costing more than typical revenue, assess the rest through their own cash-flow plans, and budget for a year of the below-target shortfall.

Table 6. Four-week inspection plan

Timing and owner	Action	Decision condition
Week 1 (Operator)	Pool ten candidate homes in the priority groups; check one-night stays and price	Inspect homes meeting both first, in group order
Weeks 2–3 (Operator & Analyst)	Get written landlord consent; check that no owners' corporation rule bans stays under 28 days (Consumer Affairs Victoria, n.d.); negotiate rent; get cost quotes	Replace homes that fail any check
Week 4 (Client)	Cash-flow plan: quotes and Victoria's 7.5% short-stay levy (State Revenue Office Victoria, n.d.) against typical and below-target revenue (Recommendation 3)	Sign only if the plan covers setup costs, required return and a year of below-target loss
First 30 days (Operator)	Record paid nights, stay length, workload, net cash flow and reviews	Expand only if results match the plan

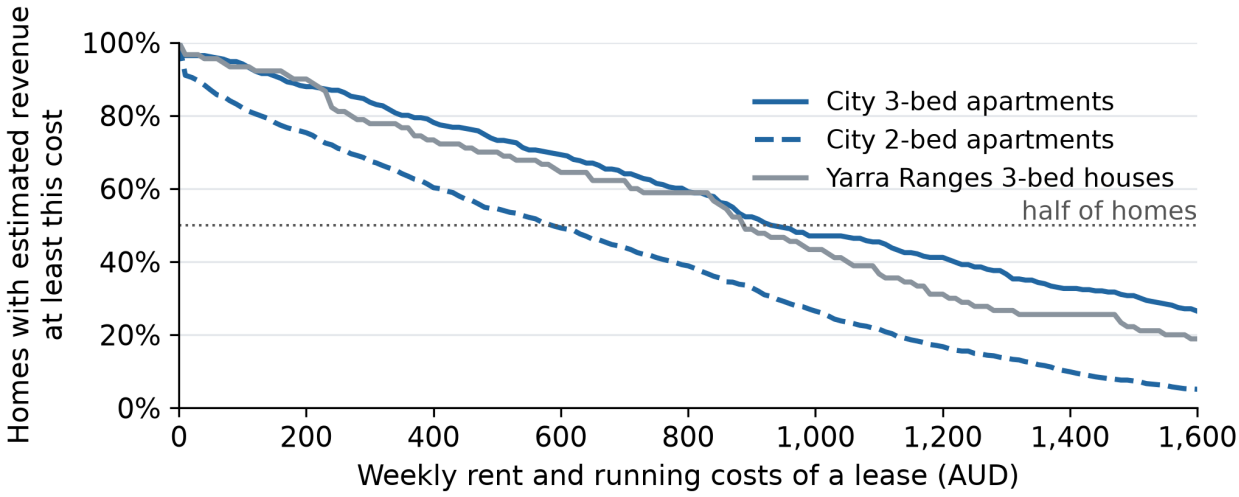


Figure 6. Share of each priority group's homes whose estimated weekly revenue reached a given weekly cost.

Worked example (illustrative costs). A City two-bedroom apartment costing **AUD 620** a week in rent and running costs exceeds its group's typical AUD 590 and is dropped. A Yarra Ranges house costing **AUD 700** clears its typical AUD 890; earning like a below-target home (AUD 600), it would lose AUD 100 a week, so the buffer needs AUD 5,200 a year.

7. Limitations and Future Work

- **Reviews are not bookings, profit or satisfaction.** Inside Airbnb's 50% review rate sits within published estimates of 30.5%–72% (Inside Airbnb, n.d.), so dollar figures are rough; activity also fell in the later year (Figure 5). Ratings are lifetime averages near five stars.
- **One snapshot of surviving homes.** Homes that left Airbnb are missing, revenue uses June–July prices, and new, unpriced and outer-area listings are not covered.
- **Short stays inflate reviews and revenue.** A one-night minimum allows more stays for the same nights, so part of its effect is built into the target, and counting each stay as three nights may overstate revenue.
- **Associations, not causes.** Settings may respond to earlier performance, and we chose the inputs, rule and priority groups after seeing all homes, so the tests may overstate the gain.
- **No rent, cost or regulation data.** Recommendation 3 relies on the client's own quotes.

Next, collect rent and cost quotes, track paid nights and net cash flow to test whether review activity identifies good businesses, analyse review text in the priority groups, and recheck the groups, model and two-setting rule on the next snapshot.

8. References

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AI acknowledgement

ChatGPT, Gemini and Claude Code assisted with writing and coding; Eric Huang and Maksym Xu reviewed the outputs.

Author contributions

Maksym Xu led data preparation, modelling and validation; Eric Huang led report drafting and review. Earlier team contributions are recorded in the repository guide.

Appendix A. Model Details

Group median prices and stand-ins for missing minimum stays use only homes each model was built on. In Table A1, each odds ratio (from the input's average weight over the 50 fits) multiplies the odds of being busy (busy homes per quiet home); above 1 raises the chance, below 1 lowers it.

Table A1. Odds ratios across 50 fits

Input (compared with)	Odds ratio	Range across fits
Accepts one-night stays (no)	3.27	2.53–4.05
Ten more amenities	1.27	1.15–1.41
Price 10% above similar homes	0.88	0.83–0.94
Council area (City of Melbourne)		
Port Phillip	0.50	0.33–0.66
Stonnington	0.48	0.26–0.68
Yarra	0.83	0.59–1.11
Yarra Ranges	1.22	0.76–1.97
Merri-bek	0.75	0.36–1.19
House or townhouse (apartment)	0.78	0.57–1.30
Two bedrooms (one)	1.16	0.89–1.41
Three bedrooms (one)	1.31	0.86–1.84

The ranges show stability across fits, not confidence intervals; those including 1 (for example, Yarra Ranges) leave the direction uncertain. Across all 14 groups, the model ranked a random busy home above a random quiet one 71% of the time, against 60% for group rates (AUC 0.71 and 0.60). Treating same-host homes as related widens margins of error to 1.1–2.6 times ordinary ones. Inputs overlap little (variance inflation factors below 3). An ordinary test of curved price and amenity effects suggests slight curvature ($p = 0.03$; amenities alone, $p = 0.011$); treating same-host homes as related makes this unclear (amenities, $p = 0.17$). Full evaluation tables are in the repository (Huang & Xu, 2026).

Table A2. Hits and misses: top quarter picked across all 14 groups

Selection	Busy	Below target	Total
Picked by model	423	546	969
Not picked	558	2,346	2,904

Each home was scored by a model built without its host (five folds of hosts). Of the 969 picked, 43.7% were busy (precision), against 25.3% overall, and they included 43.1% of busy homes (recall). Accuracy (71.5%) misleads: calling every home "below target" scores 74.7%.